VID MVP

Scalable School ERP – MVP Version

Virtual Identification System

MVP GOAL

Build Prototype

Create a functional web application that demonstrates the core capabilities of the VID platform

Validate Architecture

Test the multi-tenant architecture and modular workspace design under real-world conditions

Enable Workflows

Allow real users to perform end-to-end tasks across different roles and workspaces

WHY MVP?

Reduce Complexity

Focus only on essential features. Avoid over-engineering and deliver a working product faster

Faster Development

Ship quickly with a lean feature set. Validate assumptions before committing to full build-out

Early Feedback

Get real user input early. Identify pain points and refine the product before scaling up

Iterative Improvement

Build in cycles. Each iteration adds value and moves the product closer to the full vision

MVP SCOPE

INCLUDED

- ✓ Authentication & Login System
- ✓ Role-Based Access Control (RBAC)
- ✓ Academic System Management
- ✓ Student Management Module
- ✓ Examination & Grading System
- ✓ Fees & Payment Tracking
- ✓ Timetable & Scheduling

EXCLUDED

- ✗ AI Features (YANTRA Advanced)
- ✗ Transport Management
- ✗ Hostel Management
- ✗ Library Management
- ✗ Advanced Analytics Dashboard

SYSTEM ARCHITECTURE

MVP Multi-Tenant Design

Multi-Tenant

Basic tenant isolation per institution

API Backend

RESTful API layer for all operations

Modular Workspaces

Independent feature modules per role

Each institution operates in an isolated tenant space. The API backend serves all tenants uniformly. Workspaces are loaded based on the user's role and permissions.

USER ROLES

Super Admin

Platform-level control

Manage all institutions

System configuration

Global settings

Institution Admin

Institution management

User provisioning

Fee & exam setup

Reports & oversight

Faculty

Class management

Attendance tracking

Marks entry

Timetable view

Student

Personal dashboard

View attendance

Check marks & exams

Fee status

CORE WORKSPACES

MVP Modules Available in the System

- Academic
- Student
- Faculty
- Examination
- Payment
- Timetable
- Common Workspace
- Admin Dashboard

Each workspace is a self-contained module with its own navigation, data views, and role-based access controls.

KEY FEATURES

Login System

Secure authentication with username and password. JWT-based session management for all users.

Role Dashboards

Personalized dashboards for each role. Relevant data and actions presented based on permissions.

Attendance

Track and record student attendance. Faculty can mark daily attendance with quick actions.

Marks Entry

Enter and manage exam marks. Students can view their results through their personal dashboard.

Fee Tracking

Monitor fee payments and dues. Admins can generate fee reports and track outstanding balances.

Timetable

View class schedules and manage timetables. Faculty and students see their assigned schedules.

USER FLOW

ADMIN FLOW

Login → Setup → Assign → Manage

FACULTY FLOW

Login → Classes → Attendance → Marks

STUDENT FLOW

Login → Dashboard → View Data

DATA MODEL

TENANT ISOLATION

institution_id

Every record is tagged with an institution identifier. This ensures complete data isolation between schools. Queries are automatically scoped to the user's institution.

CORE TABLES

Table	Purpose
users	Authentication & roles
students	Student profiles & data
exams	Exams & results
fees	Fee structure & payments

UI/UX DESIGN

Responsive Web App

Works seamlessly across desktop, tablet, and mobile devices with adaptive layouts

Dashboard Navigation

Role-based dashboards show relevant data at a glance with quick-action cards

Workspace Sidebar

Collapsible sidebar organizes all modules. Users switch contexts without losing their place

SECURITY

JWT Authentication

Stateless token-based auth system. Secure login with encrypted tokens and automatic expiration

Role-Based Permissions

Granular access control. Each role sees only authorized features and data endpoints

Tenant Isolation

Institution-scoped data queries ensure schools never access each other's information

PERFORMANCE

Fast APIs

Optimized backend endpoints deliver sub-second response times. Efficient query patterns minimize database load and ensure smooth user interactions even under concurrent usage

Indexed DB Queries

Strategic database indexing on tenant_id, user_id, and frequently queried fields ensures rapid data retrieval and scalable performance as the dataset grows

DEVELOPMENT PLAN

01
Backend
API, DB schema, auth, RBAC

02
Frontend
UI components, dashboards, forms

03
Integration
Connect frontend to backend APIs

04
Launch
Deploy, test, and release MVP

SUCCESS METRICS

100%

Working Login

All four user roles can authenticate and access their respective dashboards securely

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Functional Workflows

All MVP workspaces are operational with complete end-to-end user journeys

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Multi-Role Usability

Super Admin, Institution Admin, Faculty, and Student roles all have functional access

FUTURE ROADMAP

Add AI (YANTRA)

Integrate intelligent assistants for automated grading, predictive analytics, and personalized learning recommendations

More Workspaces

Expand with Transport, Hostel, Library, and Advanced Analytics modules for complete campus management

Mobile App

Native mobile applications for iOS and Android to extend access beyond the desktop web experience

CONCLUSION

MVP validates full VID vision

Ready for scaling